MAGMATOMIX/UMD 3.0

from molecular-dynamics runs to properties, with their errors

ab initio and machine-learning molecular dynamics

VASP
ABINIT
CP2K
LAMMPS
QBox

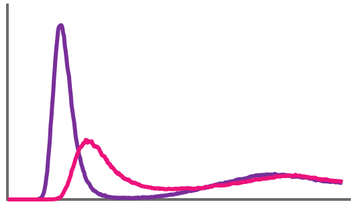
.umd.dat

one format for all codes

check_umd.py
14 quality checks

one command : umd_workflow.py

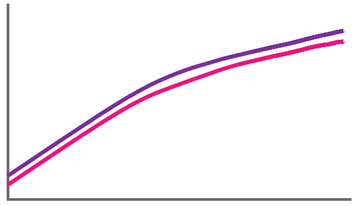
STRUCTURE



g(r) of Si–O, Mg–O

g(r), bonds, coordination
speciation, angles
lifetimes of species
liquid / vapor
rotations of molecules

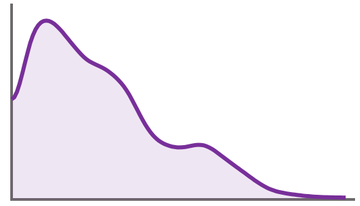
TRANSPORT



MSD of O, Si (log–log)

diffusion and its tensor
diffusion of species
viscosity
velocity autocorrelation

THERMODYNAMICS



vDOS

averages, heat capacity
vibrational spectra
2PT entropy
free energy (integration)

any cell

NVT and NPT runs,
non-orthogonal cells

± errors

on every statistical
quantity, calibrated

validated

38 cases, synthetic runs
with exact answers

6× faster

parallel workers,
memory bounded

Schools: Bucharest, April 2027

github.com/rcaracas/MAGMATOMIX-UMD_package

next, 3.x: towards 10^9 atoms